Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/16/2019
PI Name:	Espinosa, Araceli Ph.D.		
Project Title:	The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and derived-Oligodendrocytes		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	araceli@ucla.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-1747
Organization Name:	University of California, Los Angeles		
PI Address 1:	635 Charles E Young Dr South, Suite 260G/NRB		
PI Address 2:			
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-7332	**Congressional District:**	33
Comments:	araceli@g.ucla.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 10/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 2/4/2020) NOTE: Extended to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19) NOTE: Extended to 11/30/2019 per F. Hernandez/ARC (Ed., 10/24/18) NOTE: Extended to 10/31/2018 per F. Hernandez/ARC (Ed., 11/2/17)		
Key Personnel Changes/Previous PI:	August 2019 report: no changes have occurred.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Recent data from NASA shows that astronauts present vision, physiological, and anatomical changes as well as intracranial pressure (ICP) and visual impairment intracranial pressure (VIIP) (more recently known as Spaceflight Associated Neuro-ocular Syndrome (SANS)) after long-duration space missions. These changes are unlike any clinical entity previously seen. Yet, the cause(s) of chronically elevated ICP/VIIP and potential consequences are unknown. The main goal of this International Space Station (ISS) spaceflight proposal is to obtain reference data on the effects of real microgravity (0G) to validate and expand our recent data obtained using simulated microgravity (sim-μG). We want to assess the impact of real microgravity on the cell cycle of human neural stem cells and derived oligodendrocytes (OL) that are the myelin forming cells in the central nervous system. We now have evidence that sim-μG increases the number of neural progenitors and OL progenitor (OLP) via cell proliferation.
Here we postulate that microgravity-associated VIIP and altered ophthalmic anatomy (papilledema, choroidal folds, etc.) as well as a potential microgravity-associated ICP without visual impairment might be at least in part due to increased cell proliferation in the brain.

Aim I. To test the hypothesis that neural stem cells (NSC) and OL progenitors exposed to real microgravity will proliferate faster and will give rise to more progenies than cells grown in 1G.

Aim II. To test the hypothesis that in real microgravity NSC and OL produce and secrete one or more molecules that may act as regulators of cell proliferation.

Significance. Neural cells possess intrinsic plasticity that can be modulated by the environment and their response to real microgravity may confirm and even reveal new properties of these cells triggered by it. We also propose to examine molecules secreted by these cells while in space in order to identify potentially unique regulators of cell proliferation. It is known that secretome proteins play a key role in cell signaling and cell migration. These molecules may also hold a potential key for the modulation of normal cell proliferation for future therapeutic interventions for the central nervous system (CNS).

Approach. Human NSC and OLP will be plated using our proprietary specific culture media in self-contained hardware designed for 0G (Astrium). Control cells will be maintained in 1G.

Ground based experiments. We will analyze the effects of real microgravity on NSC and OL cell cycle via immunocytochemistry. After landing and recovery of the samples, the conditioned culture medium will be collected, maintained at 4°C, and transported to our laboratory to further characterize the secretome of both cell types maintained in real microgravity. Our stem cell culture medium as well as the oligodendrocyte specification medium are both serum-free and contain few proteins, properties that will facilitate the secretome analysis. The characterization of the secretome will include identification of proteins and small molecules that may be produced and secreted in real microgravity only.

Innovation. This is the first study on the impact of real microgravity on human neural cells. Our data will provide valuable information to better understand the gravitational epigenetic modulation of human neural cells.

Benefits for Humanity. Confirming that real microgravity increases cell proliferation will move us forward toward a better understanding on the causes of ICP and VIIP in astronauts. Moreover, since long-term missions in space are imminent, the identification of molecules that increase cell proliferation might lead to the design of potential countermeasures to prevent or alleviate the negative effects observed in astronauts exposed to long-term missions in space. Myelin is responsible for the rapid conduction of electrochemical messages to and from the brain. Since there is no cure for myelin disorders our data might also lead to novel therapeutic modalities for myelin repair.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

To date, there is no cure for developmental disabilities, neurodegenerative disorders, myelin loss, or cancer. We expect that our work described in the current proposal will generate cures that are of paramount importance in two different but complementary fields:
1) Drug discovery: The first product will be the first molecule that would have been identified, isolated, and tested to be responsible for increased cell proliferation. This will open the door for a second product (or series of products) under this category that would be an antibody or similar molecule(s) that would void/neutralize the effects of the first molecule on cell proliferation. This second product will be further studied as an anti-cancer agent via prevention of cell proliferation. If there is more than one relevant secreted molecule a similar process would be followed. Once we have data showing that we have interesting molecules, we will submit an invention report to the Office of Intellectual Property at UCLA (University of California, Los Angeles). This office supports scientists by contacting potentially interested companies and helping disseminate information on technologies available for licensing. In some cases UCLA technologies are commercialized by large companies. Alternatively, some entrepreneurs-inventors form successful startups based on their own UCLA technology.

2) Generation of cells for cell replacement therapies: The presence of functional myelin is essential for all our normal daily activities. Our overall goal is to investigate the prospects of cell replacement therapies for patients that require normal OL and myelin to restore proper CNS function. We contemplate to be able to apply this approach to propagate OLPs also known as (OPCs) and make them available to treat the spectrum of myelin disorders, from white matter damage in the premature neonate and infants suffering of autism or maternal prenatal infection, to the patient with leukodystrophies, multiple sclerosis, psychosis/schizophrenia, and trauma. We have previously demonstrated that OLPs are cells that, once grafted, migrate, integrate, and interact with the host CNS. Validating the functionality of OLPs exposed to real μG will open a new era on the ways we conceive cell replacement therapies and CNS repair. Moreover, trauma to the brain and spinal cord involves myelin loss and glial and neuronal cell death; thus, data generated by the current study will provide insights on producing cells to be transplanted into the CNS of individuals suffering from trauma.

Prior to commercializing the generation of cell populations in microgravity, it is essential to confirm that microgravity-exposed cells are fully functional and safe; by biochemical and pre-clinical neural transplantation proposed in this grant application we will generate the necessary evidence. Once this step will have been accomplished, we'll contact specific companies that might provide us with the revenue necessary to fund the scale-up process to generate human induced pluripotent stem cell (hiPS) derived oligodendrocytes progenitors (OLP). By then we hope to have created strong partnership with the industry in order to be able to generate cells to overcome neural degeneration and

	developmental disabilities.
Task Progress:	The BioScience-4 mission launched on December 5, 2018 onboard Space X-16. Pre-flight operations At UCLA: Labels for cultures and reagents/growth factors' aliquots were prepared and printed at the Principal Investigator's (PI's) laboratory. Two cases containing items that were essential for flight implementation were sent to the Cape. Culture media was prepared and sent frozen. Stocks of frozen cells and two sets of growing cultures were also sent to Kennedy Space Center (KSC). PI activities at the Kennedy Space Center: The PI worked with STaARS from scratch in preparation for flight integration because the STaARs team members were different from those participating in the ERD and Experimental Verification Tests (EVT). The Airbus-Kiwi team joined us at KSC for flight implementation and handover was timely and without incident. Cells stayed 39.3 days in space followed by splash-down and transfer to the PI laboratory. The PI had everything ready and de-integration of the flight was performed. Human oligodendrocytes and neural stem cells (NSCs) were recovered from the flying devices and seeded for further studies. The secretome was collected and placed in their respective tubes for both metabolomics and proteomics. Cultures were fed to evaluate their viability and health status. Oxygen consumption and glycolysis (anaerobic respiration) were determined and data collected. NSCs recovered faster from spaceflight. Oligodendrocyte progenitors (OLPs) did not respond well to extended and repeated exposure to room temperature. Thus, OLPs were recovered in scarce numbers. In addition, events inherent to this specific spaceflight led to the need of two supplementary experiments. The PI performed the ground control implementation and supplementary experiments in a concomitant manner. This is an ongoing study and the secretome has been processed to examine the protein contents and metabolites. Data analysis will start in September, 2019. We gladly report that NSCs coming back from space preserved their stemness including cell renewal and potential to give rise to neurons. OLPs that survived after spaceflight also gave rise to healthy progenies. These OLPs generated by OLPs flown to space present characteristics very close to those shown by ground control OL. We will be submitting a paper to a peer-reviewed journal in the next few months [Ed. note 9/30/19: article submitted to Applied Sciences, published 9/27/2019, and listed in Bibliography section below]. The following will be presented later this year at the 35th annual meeting of the American Society for Gravitational & Space Research in Denver: C. Cepeda, L.Vergnes, L, N. Carpo, S. Shaka, M.J. Schiber L. A. Bentolia, F. Karouia, A. Espinosa-Jeffrey. "How does space flight influence the biology of human induced pluripotent-derived neural stem cells?"
Bibliography Type:	Description: (Last Updated: 02/07/2024)
Abstracts for Journals and Proceedings	Cepeda C, Vergnes L, Carpo N, Shaka S, Schiber MJ, Bentolia LA, Karouia F, Espinosa-Jeffrey A. "Human Neural Stem Cells Flown to Space Survive, Preserve Their Pluripotency and Give Rise to Young Neurons." Presented at the 2019 International Space Station Research and Development Conference (ISSRDC), Atlanta, GA, July 29-August 1, 2019. 2019 International Space Station Research and Development Conference (ISSRDC), Atlanta, GA, July 29-August 1, 2019. , Aug-2019
Articles in Peer-reviewed Journals	Cepeda C, Vergnes L, Carpo N, Schibler MJ, Bentolia LA, Karouia F, Espinosa-Jeffrey A. "Human neural stem cells flown into space proliferate and generate young neurons." Applied Sciences. 2019;9(19):4042. Published: 27 September 2019. https://doi.org/10.3390/app9194042 [Note originally reported in August 2019 as "submitted"] , Sep-2019